script.js

Online Quiz Application - logic (JavaScript)

```
/* ---------- Question bank (4 categories x 10 questions) ---------- */
const QUIZ_DATA = {
  "General Knowledge": {
    desc: "Everyday facts about the world", time: 180,
    questions: [
      { q: "Which planet is known as the Red Planet?", o: ["Venus", "Mars", "Jupiter", "Mercury"], a: 1 },
      { q: "What is the capital of Australia?", o: ["Sydney", "Melbourne", "Canberra", "Perth"], a: 2 },
      { q: "How many continents are there on Earth?", o: ["5", "6", "7", "8"], a: 2 },
      { q: "Which is the largest ocean?", o: ["Atlantic", "Indian", "Arctic", "Pacific"], a: 3 },
      { q: "Who wrote the national anthem of India?", o: ["Bankim Chandra Chatterjee", "Rabindranath Tagore", "Sarojini Naidu", "Subramania Bharati"], a: 1 },
      { q: "Which gas do plants absorb from the atmosphere?", o: ["Oxygen", "Nitrogen", "Carbon dioxide", "Hydrogen"], a: 2 },
      { q: "What is the currency of Japan?", o: ["Yuan", "Won", "Yen", "Ringgit"], a: 2 },
      { q: "Which is the longest river in India (within India)?", o: ["Godavari", "Ganga", "Yamuna", "Krishna"], a: 1 },
      { q: "How many days are there in a leap year?", o: ["365", "366", "364", "367"], a: 1 },
      { q: "Which animal is called the 'Ship of the Desert'?", o: ["Horse", "Elephant", "Camel", "Donkey"], a: 2 }
    ]
  },
  "Science": {
    desc: "Physics, chemistry and biology basics", time: 180,
    questions: [
      { q: "What is the chemical symbol for water?", o: ["H2O", "CO2", "O2", "NaCl"], a: 0 },
      { q: "Which part of the cell contains genetic material?", o: ["Cytoplasm", "Nucleus", "Ribosome", "Membrane"], a: 1 },
      { q: "What is the SI unit of force?", o: ["Joule", "Watt", "Newton", "Pascal"], a: 2 },
      { q: "Which gas is most abundant in Earth's atmosphere?", o: ["Oxygen", "Nitrogen", "Argon", "Carbon dioxide"], a: 1 },
      { q: "Speed of light in vacuum is approximately:", o: ["3 x 10^6 m/s", "3 x 10^8 m/s", "3 x 10^10 m/s", "3 x 10^5 m/s"], a: 1 },
      { q: "Which vitamin is produced when skin is exposed to sunlight?", o: ["Vitamin A", "Vitamin B12", "Vitamin C", "Vitamin D"], a: 3 },
      { q: "The pH of pure water at 25°C is:", o: ["0", "7", "14", "1"], a: 1 },
      { q: "Which organ pumps blood through the human body?", o: ["Lungs", "Liver", "Heart", "Kidney"], a: 2 },
      { q: "What is the atomic number of carbon?", o: ["4", "6", "8", "12"], a: 1 },
      { q: "Which planet has the most moons (as commonly taught)?", o: ["Earth", "Mars", "Saturn", "Mercury"], a: 2 }
    ]
  },
  "Programming": {
    desc: "Web and coding fundamentals", time: 150,
    questions: [
      { q: "What does HTML stand for?", o: ["Hyper Text Markup Language", "High Tech Modern Language", "Hyperlinks and Text Markup Language", "Home Tool Markup Language"], a: 0 },
      { q: "Which language adds interactivity to web pages?", o: ["CSS", "HTML", "JavaScript", "XML"], a: 2 },
      { q: "Which CSS property changes the text colour?", o: ["font-style", "color", "background", "text-align"], a: 1 },
      { q: "Which method stores data in the browser that persists after closing it?", o: ["sessionStorage", "Cookies only", "localStorage", "Cache API only"], a: 2 },
      { q: "What does the === operator do in JavaScript?", o: ["Assigns a value", "Compares value only", "Compares value and type", "Declares a constant"], a: 2 },
      { q: "Which HTML tag creates a hyperlink?", o: ["<link>", "<a>", "<href>", "<url>"], a: 1 },
      { q: "Which data structure works on the LIFO principle?", o: ["Queue", "Stack", "Array", "Tree"], a: 1 },
      { q: "What does CSS stand for?", o: ["Creative Style Sheets", "Computer Style Sheets", "Cascading Style Sheets", "Colorful Style Sheets"], a: 2 },
      { q: "Which function converts a JSON string into an object?", o: ["JSON.stringify()", "JSON.parse()", "JSON.object()", "JSON.convert()"], a: 1 },
      { q: "Which symbol starts a single-line comment in JavaScript?", o: ["#", "<!--", "//", "**"], a: 2 }
    ]
  },
  "Mathematics": {
    desc: "Arithmetic, algebra and geometry", time: 180,
    questions: [
```

```
      { q: "What is 15 x 12?", o: ["170", "180", "190", "160"], a: 1 },
      { q: "What is the square root of 144?", o: ["10", "11", "12", "14"], a: 2 },
      { q: "The sum of angles in a triangle is:", o: ["90°", "180°", "270°", "360°"], a: 1 },
      { q: "What is 25% of 240?", o: ["50", "60", "70", "80"], a: 1 },
      { q: "Which of these is a prime number?", o: ["21", "27", "29", "33"], a: 2 },
      { q: "Value of π (approx.) to two decimals:", o: ["3.12", "3.14", "3.41", "3.16"], a: 1 },
      { q: "Solve: 2x + 6 = 14. x = ?", o: ["3", "4", "5", "6"], a: 1 },
      { q: "Area of a rectangle 8 cm by 5 cm:", o: ["13 cm²", "26 cm²", "40 cm²", "45 cm²"], a: 2 },
      { q: "What is 7³ ?", o: ["21", "49", "243", "343"], a: 3 },
      { q: "How many sides does a hexagon have?", o: ["5", "6", "7", "8"], a: 1 }
    ]
  }
};

/* ---------- State ---------- */
const $ = id => document.getElementById(id);
const screens = ["screen-home", "screen-quiz", "screen-result"];
let current = null;     // category name
let questions = [];     // shuffled questions for this attempt
let answers = [];       // selected option index per question (null = unanswered)
let idx = 0;
let remaining = 0;
let timerId = null;

/* ---------- Helpers ---------- */
function show(id) { screens.forEach(s => $(s).classList.toggle("hidden", s !== id)); }

function shuffle(arr) {                       // Fisher-Yates
  const a = arr.slice();
  for (let i = a.length - 1; i > 0; i--) {
    const j = Math.floor(Math.random() * (i + 1));
    [a[i], a[j]] = [a[j], a[i]];
  }
  return a;
}

function fmt(s) {
  const m = String(Math.floor(s / 60)).padStart(2, "0");
  const r = String(s % 60).padStart(2, "0");
  return m + ":" + r;
}

function getBest() {
  try { return JSON.parse(localStorage.getItem("quizBestScores")) || {}; }
  catch (e) { return {}; }
}
function saveBest(cat, score, total, percent) {
  const best = getBest();
  const prev = best[cat];
  let isNew = false;
  if (!prev || score > prev.score) {
    best[cat] = { score, total, percent, date: new Date().toLocaleDateString() };
    try { localStorage.setItem("quizBestScores", JSON.stringify(best)); } catch (e) {}
    isNew = true;
  }
  return { best: best[cat], isNew };
}

/* ---------- Home screen ---------- */
function renderHome() {
  clearInterval(timerId);
  const best = getBest();
  const box = $("categories");
  box.innerHTML = "";
  Object.keys(QUIZ_DATA).forEach(name => {
    const c = QUIZ_DATA[name];
    const b = best[name];
    const btn = document.createElement("button");
    btn.className = "cat";
```

```
    btn.innerHTML = "<b></b><span class='d'></span><span class='t'></span><span class='b'></span>";
    btn.querySelector("b").textContent = name;
    btn.querySelector(".d").textContent = c.desc;
    btn.querySelector(".t").textContent = c.questions.length + " questions · " + fmt(c.time);
    btn.querySelector(".b").textContent = b ? "Best: " + b.score + "/" + b.total + " (" + b.percent + "%)"
  : "Best: not attempted";
    btn.onclick = () => startQuiz(name);
    box.appendChild(btn);
  });
  show("screen-home");
}

/* ---------- Quiz ---------- */
function startQuiz(name) {
  current = name;
  questions = shuffle(QUIZ_DATA[name].questions).map(q => {
    // also randomize option order, remembering the correct one
    const order = shuffle(q.o.map((text, i) => ({ text, correct: i === q.a })));
    return { q: q.q, o: order.map(x => x.text), a: order.findIndex(x => x.correct) };
  });
  answers = new Array(questions.length).fill(null);
  idx = 0;
  remaining = QUIZ_DATA[name].time;
  $("cat-name").textContent = name;
  show("screen-quiz");
  renderQuestion();
  updateTimer();
  clearInterval(timerId);
  timerId = setInterval(() => {
    remaining--;
    updateTimer();
    if (remaining <= 0) finishQuiz(true);   // auto-submit
  }, 1000);
}

function updateTimer() {
  const t = $("timer");
  t.textContent = fmt(Math.max(remaining, 0));
  t.classList.toggle("low", remaining <= 30);
}

function renderQuestion() {
  const q = questions[idx];
  $("q-count").textContent = "Question " + (idx + 1) + " of " + questions.length;
  $("q-text").textContent = q.q;
  $("bar").style.width = ((idx + 1) / questions.length * 100) + "%";
  const box = $("options");
  box.innerHTML = "";
  q.o.forEach((text, i) => {
    const b = document.createElement("button");
    b.className = "opt" + (answers[idx] === i ? " sel" : "");
    b.textContent = String.fromCharCode(65 + i) + ". " + text;
    b.onclick = () => { answers[idx] = i; renderQuestion(); };
    box.appendChild(b);
  });
  $("btn-prev").disabled = idx === 0;
  $("btn-next").textContent = idx === questions.length - 1 ? "Submit" : "Next";
}

$("btn-prev").onclick = () => { if (idx > 0) { idx--; renderQuestion(); } };
$("btn-next").onclick = () => {
  if (idx < questions.length - 1) { idx++; renderQuestion(); }
  else finishQuiz(false);
};

/* ---------- Result ---------- */
function finishQuiz(timedOut) {
  clearInterval(timerId);
  const total = questions.length;
```

script.js

Online Quiz Application - logic (JavaScript)

```
  const attempted = answers.filter(a => a !== null).length;
  const correct = answers.filter((a, i) => a === questions[i].a).length;
  const incorrect = attempted - correct;
  const unanswered = total - attempted;
  const percent = Math.round(correct / total * 100);
  const { best, isNew } = saveBest(current, correct, total, percent);

  $("score-text").textContent = correct + " / " + total;
  $("percent-text").textContent = percent + "% " + (timedOut ? "· Time expired, quiz auto-submitted" : "");
  $("best-text").textContent = (isNew ? "New best score! " : "Best score: ") + best.score + "/" + best.total + " (" + best.percent + "%) on " + best.date;

  const rows = [
    ["Category", current],
    ["Total questions", total],
    ["Attempted questions", attempted],
    ["Unanswered", unanswered],
    ["Correct answers", correct],
    ["Incorrect answers", incorrect],
    ["Percentage", percent + "%"],
    ["Time taken", fmt(QUIZ_DATA[current].time - Math.max(remaining, 0))]
  ];
  $("summary").innerHTML = "";
  rows.forEach(r => {
    const tr = document.createElement("tr");
    const a = document.createElement("td"); a.textContent = r[0];
    const b = document.createElement("td"); b.textContent = r[1];
    tr.append(a, b);
    $("summary").appendChild(tr);
  });

  const rev = $("review");
  rev.innerHTML = "";
  questions.forEach((q, i) => {
    const sel = answers[i];
    const ok = sel === q.a;
    const d = document.createElement("div");
    d.className = "rev" + (ok ? " right" : "");
    const p1 = document.createElement("p"); p1.innerHTML = "<b></b>"; p1.firstChild.textContent = (i + 1) + ". " + q.q;
    const p2 = document.createElement("p");
    p2.innerHTML = "Your answer: <span></span>";
    const sp = p2.querySelector("span");
    sp.textContent = sel === null ? "Not answered" : q.o[sel];
    sp.className = ok ? "tag-ok" : "tag-bad";
    d.append(p1, p2);
    if (!ok) {
      const p3 = document.createElement("p");
      p3.innerHTML = "Correct answer: <span class='tag-ok'></span>";
      p3.querySelector("span").textContent = q.o[q.a];
      d.appendChild(p3);
    }
    rev.appendChild(d);
  });
  show("screen-result");
  window.scrollTo(0, 0);
}

$("btn-restart").onclick = () => startQuiz(current);
$("btn-home").onclick = renderHome;

renderHome();
```